

GrandmaPath



- Grandma + grandpa + path
- Increases friendliness / fondness by the words 'grandma' and 'grandpa' instead of elderly or seniors

1. Who are you designing a solution for?

There is a famous Korean saying - 'It is darkest under the lamp.' In other words, you often do not realize what is right under your nose. Society is doing the same, especially in South Korea. Out of all the pandemics and global crises, the one crucial thing we sometimes forget is the 4th industrial revolution. It was a massive change with both sides of the coin. It certainly had positive effects on humanity, but what use is it if 20% of South Korea's population can not utilize it properly? The elderly are excluded from the "useful" technology; while others can promote their own well-being from this technology, the elderly are not able to.

In 2025, South Korea is expected to become a super-aged society, with more than 20% of the population being seniors over the age of 65. At the same time, South Korea has the highest elderly pedestrian death rate among the OECD countries. While the OECD countries' average number of deaths among ten thousand elderly pedestrians is three people, South Korea has 14.39. The main problem is that this number is increasing and will increase even more as the country becomes a super-aged society. And there is no actual solution to solve this. This is why I am designing a navigation system just for the elderly. As the SDG 3 goal is about healthy lives and well-being for all ages, those over 65 need to be considered. The elderly are underrepresented, as they are rarely prioritized over the disabled, such as the blind or the poor. But as South Korea has an exceptionally high elderly pedestrian death rate and certainly a lot of senior citizens, they are the ones who need the most help and attention. I firmly believe that it is society's role to take care of senior citizens, especially at a time when they are getting excluded from the rapid improvement of technology.

2. What customer need or want are you trying to meet?

There are four needs this application fulfills:

1. Finding the safest roads.
2. Making the navigation application easy to use.
3. Defining the places and modes of transportation the elderly utilize often.
4. Providing essential information that the elderly have difficulty accessing is crucial.

In modern society, where everyone is busy, speed is crucial. But is this for everyone? The answer is no. The fastest roads are not what the elderly need. Safety takes precedence over anything, and speed comes second. While the quickest path may be the optimal path for normal citizens, it is not the optimal path for the elderly.

Not only does the existing navigation application only consider speed or money, but it is extremely hard for the elderly to use. The tiny letters have too many functions – too much that something young people instinctively know is complicated for them.

The elderly also can not use current navigation systems properly, as it is hard for them to use typical applications. They need more accessible apps to control. As a result, seniors are increasingly excluded from technology's rapid development. This should not be something that increases the elderly pedestrian death rate in South Korea when it has the potential to decrease the numbers.

Currently, there is no proper navigation system suited for the elderly. They need a safe path they can easily use and a navigation system just right for them. They need a road with more lights, fewer cars, lower slopes, less number of accidents, and is safer.

We must also consider different aspects, such as places the elderly often use, modes of transportation, and information needed for them. As people age, it is harder for them to drive cars on their own. Thus, they tend to use public transportation, which is also accessible for those 65 years or older in South Korea. The problem is that, in most cases, the system is also very complicated for them. Navigation will also solve this problem by making it easy for them to navigate through intricate networks or public transportation.

Lastly, the elderly have difficulty accessing information about the places they want to go. Compared to the younger citizens, they are much slower in knowing facts about the places they are trying to go, such as shop closing or moving places. This navigation will also show this so that they can easily know.

3. What is your innovative idea? 0 / 500 words

The navigation app will help the elderly when traveling. It will reduce the high rates of elderly pedestrian accidents and deaths in South Korea and will be easy to use for them. The goal is to provide the seniors with the safest paths and customize the navigation system just for the elderly, such as different places 'starred' or other modes of transportation included and

different information included. The application's name is 'GrandmaPath,' made by adding the words grandma, grandpa, and path. The familiar words 'grandma' and 'grandpa' give the users and family of the users a sense of sympathy and fondness. In addition, red, yellow, and orange, which are the colors most easy to spot for the elderly, were used. By using this application, the hopes are that many seniors in South Korea will be able to travel along the safest paths confidently.

4. How would your idea work?

When the application GrandmaPath is clicked, the screen will consist of three lines, with one to two words in each line. Each will be bold and clear so that anyone can read the words. The first line will be 'Starred Places.' If selected, it will redirect the user to the page where they can scroll down to find their favorite places. It will include necessary places such as convenience stores, hospitals, and banks and places specifically for seniors, such as senior centers or town halls. Below each place, it will include whether the place is open or not. And if it is close to closing time or opening time, the time left will also be included. When one of the 'Starred Places' is clicked, they will be able to easily select the mode of transportation – walking, public transportation, senior scooters, and more. Here, if public transportation is selected, helpful buses such as senior buses run in some places of South Korea will be automatically selected if they have better options. Then, it will navigate them through the safest and quickest path. This will be easy for the elderly to use as it will have simple voiceovers, and the instructions will show in the middle of the page clearly.

If the second line, 'New Place,' is clicked, the elderly will be able to type or speak into their phone to select the new place they want to travel to. It will then have the same option as above. In addition, the places the elderly visit, which they visit often, will be automatically added to the 'Starred Places,' or they can simply click the star shape and add the destination.

The third line will have one word, 'Help.' If clicked, it will show many questions users may have, and it will kindly tell them the answer step by step.

The navigation will also include an SOS button at the top so that when the siren logo is clicked, two options will pop up – police or hospital. If pressed, calls for help will go to the police or the nearest open hospital depending. By this, the elderly will be able to call for help easily if they have any trouble while traveling.

5. How does your idea meet the challenge?

GrandmaPath fulfills the necessities for all types of the well-being of the elderly. There are three types of well-being: physical, mental, and social.

First of all, the physical well-being of the elderly is promoted when traveling as their safety is much more ensured than when they use normal navigation systems or do not use any navigation systems.

Secondly, the mental well-being of the elderly is promoted as well. Because the navigation is easy to use and is suited for the elderly, it assures the elderly that they will be safe and will be able to travel by themselves. Moreover, it gives them confidence that they can move by themselves and do not have to rely on other people to travel.

Last but not least, the social well-being of the elderly will also be promoted by the elderly being able to access the places many other elderly visit easily without getting lost. The 'Starred Places' will have options such as senior centers, where the elderly will be able to socialize with others. As they can easily access these places, they will visit them more and socialize.

6. How does your idea meet the customer's needs or wants?

GrandmaPath attempts to solve the high elderly pedestrian death rate in South Korea. The challenges confronted in this application are:

- Finding the safest paths.
- Making it accessible to the elderly.
- Making sure they can use it well.

First of all, the navigation finds the safest path by altering the algorithms to the skyline query. Each road will get a 'score' depending on the safety of the road, and the path that is quickest and safest will be shown. This can not be done by brute force as it takes too much time, and greedy algorithms are inaccurate. Therefore, we plan to use skyline queries to find the optimal path.

The interfaces will be designed to be easy to use, even for the elderly, and everything will provide a voiceover. It will be possible for the elderly to enter everything using their own voices instead of typing as well.

This application will be a government program. As the elderly will have a hard time downloading the application by themselves, the government should run a program so that posters including the detailed steps of how to download the application will be hung near apartments and places where the elderly visit often, such as senior centers, convenience stores, and banks. The senior centers or town halls should help the seniors download the application for them in person. In addition, there should be advertisements on social media so that the young can help download them for their parents or grandparents. As the application helps in reducing the death rate of the elderly, it should be a government program.